Juan Hernandez

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EDUCATION

Stanford University | GPA: 3.4/4.0 — **Stanford, CA**
Bachelor of Science Mechanical Engineering — *Graduation Date – June 2027*
Relevant Coursework: Engineering Thermodynamics; Intro to Fluids Engineering; Mechanics of Materials; Assistive Technology Design; Mechanical Systems Design; Dynamics Systems Vibrations and Controls

WORK EXPERIENCE

Northrop Grumman — **Los Angeles, CA**
System Integration and Test Engineer — *June - August 2025*
- Led a continuous improvement project to make next-generation test equipment backwards compatible, adapting backend programs for legacy navigation systems, improving reliability and reducing multi-hour test cycles.
- Conducted flight path simulations, thermal tests, and shock testing to verify GPS and Internal Navigation System communication in extreme environments through MATLAB data analysis.
- Created and updated operational procedures and user guides to streamline testing processes and establish consistency for receiver testing.

PROJECTS

Stanford University — **Stanford, CA**
Victorian Lantern — *March- June 2025*
- Conceptualized, designed, and fabricated a Victorian-inspired lantern within a $100 budget.
- Created an assembly design in Fusion 360, that guided an operation sequence for manufacturing.
- Balanced design aesthetics and manufacturability through low fidelity prototyping.

UC Irvine — **Irvine, CA**
RC Plane Design — *July 2022*
- Rapid prototyped various aircraft designs to explore airfoil characteristics, and establish the most effective configuration for stability and performance.
- Collaborated with a team to finalize a design and create a functional RC plane over the course of one month.
- Utilized laser cut styrofoam to construct the plane's shell, then wired a battery pack, propeller, and receiver to enable powered flight and remote-control functionality.

Stanford University — **Stanford, CA**
Lightweight Bike Frame — *March 2025*
- Created a detailed CAD model in Fusion 360 and performed FEA (Finite Element Analysis) to evaluate bending moments and stress distribution at key contact points.
- Conducted small-scale testing using a 3D printed model, comparing hand calculations with simulated material properties and stress data.
- Utilized results to write a report defending materials that meet the intersection of our design specifications.

LEADERSHIP

Stanford Axe Committee — **Stanford, CA**
Financial Officer & Director of Social and Recruitment — *September 2023 - Present*
- Managed a $33,000 budget, across an academic year and helped allocate the budget between a large-scale rally, working with multiple student organizations and external vendors, as well as food and equipment for the club.
- Led recruitment of new members, and fostered a community by planning events and establishing an active line of communication between club officers and new members.

SKILLS

MATLAB; Autodesk Fusion360; Python; Blender; Welding; Onshape; Spanish; Laser Cutting; 3D Printing; Rapid Prototyping; Design Sketching; Solid Works; Manual Mill; Manual Lathe